

WEEK 1 REPORT

1. Why ViT?

Before ViTs the standard approach for images was CNNs.

CNNs look at local neighbourhoods but ViT turns the image into a sequence of tokens and give that sequence to a transformer (like NLP). Images are split into patches and then these patches are treated as a sequence of tokens.

2. How image patchification works

It is basically a 2D image into a sequential 1D token list, so that transformers can process it like text.

It is divided into non overlapping fixed size patches.

For ex 224 x 224 image into fixed 16x16 sized patches so there would be $14 \times 14 = 196$ patches

3. Patch embeddings

A transformer cannot directly understand a square of pixels. So each patch is converted into a vector of numbers called an embedding.

4. Positional embeddings

But now the transformer needs to know where each patch came from so for this there is positional embeddings.

5. CLS token

CLS does not represent a physical part of the image. It is a special token used to collect information about the whole image.

6. ViT Transformer Arch

The image tokens are passed through several Transformer blocks. Each block uses self-attention to let tokens exchange information. An MLP then processes the tokens further. Layer normalization and residual connections are used in these blocks.

7. CLS vs Patch Tokens

The CLS token gives one representation for the whole image. The patch tokens give representations for individual image regions. Therefore, CLS contains global information, while patch tokens retain spatial information

8. What is DINOv2?

DINOv2 is a pretrained Vision Transformer model used to extract image features. It is trained using images without requiring a label for every image. In this project, we use it to obtain useful representations of DESI galaxy images.

9. What are DINOv2 Outputs?

DINOv2 gives a CLS token and a set of patch tokens.

The CLS token represents the complete image.

Each patch token represents one image patch.

The number of features in each token depends on the DINOv2 model used.

10. How DINOv2 will be Used

We pass DESI galaxy images through the pretrained DINOv2 model.

We extract the CLS and patch tokens from the model. We then inspect their dimensions and visualize the patch representations.

This helps us understand what features the model extracts from galaxy images.